Darke Theory: A Pre-Geometric Bridge Underlying Quantum Gravity and General Relativity

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Abstract

This paper develops Darke Theory as a pre-geometric framework underlying both the regime ordinarily described quantum mechanically and the regime ordinarily described relativistically. Its central claim is that spacetime, geometry, and physical law are not primitive, but become admissible only once coherence exceeds a critical threshold sufficient for stable closure. On this view, quantum mechanics and general relativity are not rival primitive foundations, but threshold-separated regimes of one deeper structure. Below the coherence threshold, geometric notions such as distance, duration, locality, and curvature are inapplicable. Above that threshold, geometry becomes admissible and an effective Lorentzian curvature regime appears.

Within this architecture, the relativistic closure–curvature bridge provides an explicit post-threshold field system linking metric, coherence, and matter, together with a theorem chain establishing Lorentzian regime emergence, causal hyperbolicity, matter as persistent topological coherence, gravitational consistency, weak-field reduction, and vacuum stability. The DTFE sector then supplies the first concrete matter-sector candidate for that bridge: a frozen nonlinear parent action with a live topological branch and strong classical validation.

The new result of the present paper is the derivation of the matter–geometry seam for the first concrete DTFE matter candidate in the post-threshold regime. Specifically, it identifies the canonical induced bridge matter field obtained from a nontrivial DTFE branch, proves transfer of topological nontriviality into the bridge sector, proves finite bridge energy under explicit post-threshold regularity hypotheses, derives the corresponding induced bridge stress tensor, and shows that the induced sector enters the post-threshold bridge equation as a finite persistent curvature source. The result is therefore a derived concrete matter-sector bridge inside a pre-geometric unification architecture.

1 Introduction

The problem of quantum gravity is usually posed as a confrontation between two successful but structurally incompatible descriptions. General relativity presupposes a geometric spacetime and treats gravity as curvature of that spacetime. Quantum mechanics presupposes a formal state space with unitary evolution and permits correlation structures that do not sit naturally inside a primitive geometric ontology. Standard programmes therefore attempt to quantise gravity, geometrise quantum structure, or construct a more encompassing formalism in which both appear as limiting cases.

Darke Theory begins from a different boundary. It does not start with spacetime, metric distance, temporal succession, or already-furnished law. It starts from a pre-geometric regime of relational coherence and asks under what conditions stable geometric reference becomes admissible at all. In this framework, geometry is not a primitive stage on which physics unfolds. Geometry becomes admissible only after coherence exceeds a threshold sufficient for stable closure. Matter is not primitive substance but persistent topological coherence. Gravity is not fundamental interaction in the usual sense, but the geometric projection of coherence gradients once geometry has become admissible.

If spacetime is emergent, the task is not to force quantum states into a prior geometry. The task is to show how one deeper pre-geometric structure gives rise both to the near-threshold

regime ordinarily described quantum mechanically and to the post-threshold regime ordinarily described relativistically. That is the setting of the present paper.

The architecture relevant to the present bridge question is now explicit at four levels:

(i) a pre-geometric ontological foundation in which both quantum and relativistic regimes are downstream of coherence closure;

(ii) a post-threshold relativistic bridge with explicit metric, coherence, and matter sectors;

(iii) a first executable field-theoretic matter sector, DTFE, that may furnish the first concrete realisation of post-threshold matter within that bridge;

(iv) a derived transfer showing how a nontrivial DTFE branch induces a finite-energy nontrivial bridge matter sector and enters the post-threshold curvature source under explicit regularity assumptions.

The aim of the paper is precise: to identify the point at which the matter–geometry seam becomes materially derivable, and to show that the first concrete DTFE matter candidate crosses that seam in derived form under explicit post-threshold regularity hypotheses.

2 Claim boundary and novelty

The present paper stands on earlier Darke Theory work at four points:

(a) the pre-geometric threshold architecture in which geometry becomes admissible only after closure threshold crossing;

(b) the identification of the near-threshold unresolved-coherence regime with the regime ordinarily described quantum mechanically;

(c) the relativistic closure–curvature bridge with primitive fields $(g_{\mu\nu}, \Phi, \psi)$ and its theorem chain establishing Lorentzian regime emergence, causal hyperbolicity, matter as persistent topological coherence, gravitational consistency, weak-field reduction, and vacuum stability;

(d) the DTFE parent-action sector with a live topological branch and strong classical validation.

Against that background, the present paper contributes the following new results.

Proposition 2.1 (Current paper contribution)**.** *Under explicit post-threshold regularity hypotheses, the present paper establishes:*

(a) a canonical induced bridge field obtained directly from a nontrivial DTFE branch;

(b) a topological transfer statement identifying the induced bridge topological charge with the DTFE degree;

(c) a finite-energy transfer theorem for the induced bridge field;

(d) an induced bridge stress tensor for that field;

(e) a derived curvature-sourcing statement showing that the induced DTFE sector enters the post-threshold bridge equation as a finite persistent matter source.

Remark 2.2. *The burden of the paper is therefore local and exact. It concerns the first concrete DTFE matter candidate and the derivation of its passage into the post-threshold bridge sector.*

3 Pre-geometric foundation

Darke Theory begins from a strict ontological boundary: spacetime, geometry, and physical law are not primitive. They are emergent features of a deeper pre-geometric regime of relational coherence. A dimensionless coherence functional $\Phi[C]$ is defined over relational configuration space C, and geometry is admissible if and only if coherence exceeds a critical threshold:

$$\exists\, G \iff \Phi[C] > \Phi_{\text{crit}}. \tag{1}$$

In the closed regime, coherence is locally stationary,

$$\delta\Phi = 0, \tag{2}$$

allowing persistent reference, structural memory, and the emergence of classical spacetime dynamics.

The Planck scale is reinterpreted not as a smallest unit of spacetime, but as the regime boundary at which geometric description first becomes admissible. Below that boundary, length, duration, locality, and curvature are undefined as geometric notions.

Remark 3.1. *If geometry is presupposed at the base, then the problem has already been framed inside the very regime whose emergence is supposed to be explained.*

4 Quantum regime below closure threshold

Within this framework, Darke Theory does not alter the operational formalism of quantum mechanics in the present paper. It alters the ontology assigned to that regime. Quantum states are interpreted as pre-geometric coherence structures that have not yet achieved closure sufficient to define stable spacetime relations. Superposition is therefore read as unresolved relational possibility rather than contradictory physical actuality.

Measurement is interpreted as structural resolution. When coherence commits to a closed configuration, geometric reference becomes admissible. Because closure has not yet occurred prior to such commitment, unitary evolution and reversibility are preserved at the quantum level in the ordinary formal treatment.

This perspective also offers an ontological reading of Bell-type nonlocality. Quantum correlations are established prior to the emergence of spacetime locality and therefore do not need to be generated by signalling across an already-admissible geometric separation. Once closure occurs, those correlations are inherited by geometric degrees of freedom without the introduction of superluminal signalling or preferred frames.

The role of this section is architectural. It places the regime ordinarily described quantum mechanically on the pre-geometric side of the threshold and thereby defines the bridge problem to be solved on the relativistic side.

5 Post-threshold relativistic bridge

The relativistic closure–curvature bridge provides the post-threshold field system in which effective geometry, matter, and curvature become explicit. The primitive post-threshold fields are

$$(g_{\mu\nu}, \Phi, \psi), \tag{3}$$

where $g_{\mu\nu}$ is a smooth symmetric nondegenerate tensor, Φ is a coherence-density field, and ψ is a concrete closure-matter field.

The bridge concerns only the post-threshold regime

$$\Phi(x) > \Phi_{\text{crit}} \qquad \forall x \in M. \tag{4}$$

Geometry is admissible only there. A geometry-activation coupling

$$F(\Phi) > 0 \tag{5}$$

enters the metric sector, and the bridge is built under explicit positivity, hyperbolicity, and admissibility conditions.

In full form, the imported bridge theorem chain establishes:

1. geometry becomes admissible only above a closure threshold;
2. admissibility is represented by a real functional on the field space;
3. curvature is sourced by coherence gradients, matter sectors, and admissibility stress;
4. causal structure arises from hyperbolicity and is physically filtered by closure balance;
5. stable topological coherence sectors act as matter-energy concentrations;
6. the coupled metric–coherence–matter system is gravitationally consistent on shell;
7. the post-threshold vacuum is linearly stable under explicit positivity conditions;
8. a homogeneous cosmological sector exists without taking the standard Friedmann equation as primitive.

The imported bridge theorem states that if $(g_{\mu\nu}, \Phi, \psi)$ satisfy post-threshold closure, geometric activation, positive kinetic structure, stationarity of the full bridge action, hyperbolicity, Lorentzian signature, and finite admissibility, then:

(a) the post-threshold regime admits an effective Lorentzian geometry;

(b) curvature is sourced by coherence gradients, matter sectors, and admissibility stress through

$$F(\Phi)G_{\mu\nu} = \kappa T^{\text{eff}}_{\mu\nu}; \tag{6}$$

(c) finite-energy nontrivial topological sectors of ψ act as matter-energy concentrations;

(d) physical propagation lies in the hyperbolic cone filtered by closure-balance admissibility;

(e) the coupled system is covariantly consistent on shell;

(f) the post-threshold vacuum is linearly stable under the stated positivity conditions.

This theorem chain is the post-threshold structure into which the present DTFE transfer result enters.

6 Matter as persistent closed coherence

The bridge does not treat matter as primitive inserted substance. It defines a concrete closure-matter sector by the Lagrangian

$$L_m = -\frac{1}{2} K(\Phi)\, g^{\mu\nu} \nabla_\mu \bar{\psi}\, \nabla_\nu \psi - U(\psi, \Phi), \tag{7}$$

with

$$K(\Phi) > 0, \qquad U(\psi, \Phi) = \lambda_\psi \big(|\psi|^2 - v(\Phi)^2\big)^2. \tag{8}$$

A finite-energy topological sector is defined by

$$E[\psi] < \infty, \qquad Q[\psi] \neq 0, \tag{9}$$

where $Q[\psi]$ denotes the relevant topological charge in the bridge matter class.

The imported matter theorem of the bridge states that if

$$K(\Phi) > 0, \qquad U(\psi, \Phi) \geq 0, \qquad E[\psi] < \infty, \qquad Q[\psi] \neq 0, \tag{10}$$

then ψ defines a nontrivial persistent topological coherence sector whose stress-energy acts as a persistent curvature source. Such sectors are macroscopically interpretable as matter-energy concentrations.

The problem is therefore concrete: whether the DTFE topological branch produces a field that satisfies this bridge matter criterion in derived form.

7 DTFE as the first concrete matter sector

DTFE is not the whole of Darke Theory. It is a frozen nonlinear parent-action theory that provides the first executable field-theoretic route by which post-threshold closure structure may be realised explicitly. Its frozen parent field is an $SU(2)$ field

$$U(x) \in SU(2), \qquad L_\mu := U^\dagger \partial_\mu U, \tag{11}$$

with parent action

$$S_{\mathrm{DTFE}}[U] = \int d^4x \left[-\frac{1}{2} \mathrm{Tr}(L_\mu L^\mu) + \frac{1}{16} \mathrm{Tr}([L_\mu, L_\nu][L^\mu, L^\nu]) \right]. \tag{12}$$

Under the hedgehog ansatz

$$U(x) = \cos F(R)\, \mathbf{1} + i\, \hat{x} \cdot \tau\, \sin F(R), \qquad R = |x|, \tag{13}$$

the system reduces to a live radial branch with

$$F(0) = \pi, \qquad F(\infty) = 0. \tag{14}$$

The earlier DTFE papers establish the live topological branch and its strong classical validation. The present question is whether that branch crosses into the bridge matter class in derived form.

7.1 Canonical DTFE-to-bridge induced field

The first task is to construct the induced bridge matter field from a finite-energy nontrivial DTFE branch in a form that is canonical up to the expected internal redundancy.

Setting

Let

$$M_{\text{post}} := \{x \in M : \Phi(x) > \Phi_{\text{crit}}\} \tag{15}$$

denote the post-threshold region, where geometric description is admissible. Only on this region can a geometric matter field be meaningfully defined. Below threshold, geometric matter variables are not primitive objects of the theory. Post-threshold matter must accordingly be induced from closure-bearing structure rather than assumed independently.

Let

$$U_B : M_{\text{post}} \to SU(2) \tag{16}$$

be a finite-energy nontrivial DTFE branch. The task is to construct from U_B a post-threshold bridge field ψ_{U_B} that lies in the bridge matter class and is not chosen ad hoc.

Reference doublet construction

Fix a unit reference doublet

$$\xi_0 \in \mathbb{C}^2, \qquad \|\xi_0\| = 1. \tag{17}$$

Define the associated complex doublet field

$$z_B(x) := U_B(x)\,\xi_0. \tag{18}$$

Since $U_B(x) \in SU(2)$ acts unitarily on $\mathbb{C}^2$, one has

$$\|z_B(x)\| = \|\xi_0\| = 1 \tag{19}$$

for all $x \in M_{\text{post}}$.

The bridge matter field is then defined by

$$\psi_{U_B}(x) := v(\Phi(x))\,z_B(x) = v(\Phi(x))\,U_B(x)\xi_0. \tag{20}$$

This construction is forced by the structural roles already fixed in the theory. The induced object must live only in the post-threshold regime, must carry the DTFE branch content with minimal additional structure, and must fit the bridge matter Lagrangian. The field ψ_{U_B} is the minimal post-threshold carrier of DTFE topological content into the bridge matter class.

Definition 7.1 (Canonical induced bridge field)**.** *Let $U_B : M_{\text{post}} \to SU(2)$ be a finite-energy nontrivial DTFE branch, and let $\xi_0 \in \mathbb{C}^2$ satisfy $\|\xi_0\| = 1$. The canonical induced bridge field associated to U_B is*

$$\psi_{U_B}(x) := v(\Phi(x))\,U_B(x)\xi_0. \tag{21}$$

Proposition 7.2 (Well-definedness of the induced field)**.** *The field ψ_{U_B} is well-defined on M_{post}, smooth whenever U_B and Φ are smooth, and changes only by the expected internal redundancy under change of unit reference doublet.*

Proof. Since U_B is $SU(2)$-valued and $\xi_0 \in \mathbb{C}^2$, the product $U_B\xi_0$ is a globally defined complex doublet on the post-threshold region. If U_B and Φ are smooth, then so is $\psi_{U_B} = v(\Phi)U_B\xi_0$.

Now let ξ_1 be another unit reference doublet. Since $SU(2)$ acts transitively on the unit sphere in $\mathbb{C}^2$, there exists $S \in SU(2)$ such that

$$\xi_1 = S\xi_0. \tag{22}$$

Then

$$U_B\xi_1 = U_B S\xi_0. \tag{23}$$

This changes the representative of the induced doublet but not the existence of the induced bridge sector itself. The resulting ambiguity is the expected internal redundancy of the doublet description. □

Theorem 7.3 (Canonical induced-field theorem)**.** *Let $U_B : M_{\text{post}} \to SU(2)$ be a finite-energy nontrivial DTFE branch. Then the assignment*

$$U_B \longmapsto \psi_{U_B} = v(\Phi)\, U_B\xi_0 \tag{24}$$

defines a canonical post-threshold bridge matter field up to the natural internal redundancy determined by the choice of unit reference doublet ξ_0.

Proof. Any bridge matter field induced from U_B must satisfy three conditions:

1. it must be defined only where geometry is admissible;
2. it must be constructed directly from the DTFE branch rather than added independently;
3. it must belong to the field class accepted by the bridge matter functional.

The field

$$\psi_{U_B} = v(\Phi)\, U_B\xi_0 \tag{25}$$

satisfies (i) because $v(\Phi)$ is a post-threshold coherence-controlled scale, (ii) because the only nontrivial field content is inherited directly from U_B, and (iii) because it is a coherence-scaled complex doublet of the exact kind admitted by the bridge matter sector. Any alternative induced field carrying the same content with less structure would fail either threshold fidelity or bridge compatibility.

Hence the induced field is canonical up to the expected internal redundancy associated with the choice of ξ_0. □

7.2 Derived DTFE-to-bridge transfer

The canonical induced bridge field

$$\psi_{U_B}(x) = v(\Phi(x))\, U_B(x)\xi_0, \qquad \|\xi_0\| = 1, \tag{26}$$

is now to be checked against the bridge matter criterion.

Normalised induced field and bridge charge

Because $U_B(x) \in SU(2)$ acts unitarily on $\mathbb{C}^2$, one has

$$|U_B(x)\xi_0| = 1 \tag{27}$$

for all $x \in M_{\text{post}}$. Since the bridge vacuum scale $v(\Phi)$ is positive in the post-threshold regime, the induced field has nonvanishing norm there and therefore admits a canonical normalisation

$$\widehat{\psi}_{U_B}(x) := \frac{\psi_{U_B}(x)}{|\psi_{U_B}(x)|} = U_B(x)\xi_0. \tag{28}$$

Definition 7.4 (Admissible topological setting)**.** *Let Σ be a connected smooth spatial slice of the post-threshold region on which the finite-energy DTFE branch approaches a constant vacuum value at spatial infinity. Assume further that Σ is diffeomorphic to $\mathbb{R}^3$, so that its one-point compactification Σ^+ is diffeomorphic to S^3. Under these hypotheses, the normalised induced field extends to a continuous map*

$$\widehat{\psi}_{U_B} : \Sigma^+ \to S^3. \tag{29}$$

Definition 7.5 (Induced bridge topological charge)**.** *Under the admissible topological setting above, define the induced bridge topological charge of ψ_{U_B} by*

$$Q_{\mathrm{br}}[\psi_{U_B}] := \deg(\widehat{\psi}_{U_B}) = \deg(U_B\xi_0). \tag{30}$$

Proposition 7.6 (Topological transfer lemma)**.** *Let $U_B : \Sigma^+ \to SU(2)$ be a finite-energy nontrivial DTFE branch with vacuum limit at spatial infinity, under the admissible topological setting above. Fix a unit reference doublet $\xi_0 \in \mathbb{C}^2$. Then the map*

$$\Theta_{\xi_0} : SU(2) \to S^3, \qquad \Theta_{\xi_0}(U) := U\xi_0, \tag{31}$$

is a diffeomorphism. Consequently,

$$\deg(U_B\xi_0) = \deg(U_B). \tag{32}$$

Proof. Fix coordinates so that $\xi_0 = e_1 = (1, 0)^T$. Every element of $SU(2)$ has the standard form

$$U = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1. \tag{33}$$

Then

$$Ue_1 = \begin{pmatrix} a \\ -\bar{b} \end{pmatrix}, \tag{34}$$

which ranges over the full unit sphere in $\mathbb{C}^2 \cong \mathbb{R}^4$, namely S^3. Moreover, the first column determines the second uniquely by the $SU(2)$ condition, so the map is bijective and smooth with smooth inverse. Hence Θ_{ξ_0} is a diffeomorphism. Degree is invariant under postcomposition by diffeomorphism, so

$$\deg(U_B\xi_0) = \deg(U_B).$$

□

Theorem 7.7 (Topological transfer theorem)**.** *Let U_B be a finite-energy nontrivial DTFE branch on a post-threshold spatial slice satisfying the admissible topological setting above, and let*

$$\psi_{U_B} = v(\Phi)\, U_B\xi_0 \tag{35}$$

be the canonical induced bridge field. Then

$$Q_{\mathrm{br}}[\psi_{U_B}] = \deg(U_B). \tag{36}$$

In particular, if the DTFE branch is topologically nontrivial, then

$$Q_{\mathrm{br}}[\psi_{U_B}] \neq 0. \tag{37}$$

Proof. By definition of the normalised induced field,

$$\widehat{\psi}_{U_B} = U_B\xi_0.$$

Hence

$$Q_{\mathrm{br}}[\psi_{U_B}] = \deg(\widehat{\psi}_{U_B}) = \deg(U_B\xi_0).$$

By the previous proposition, $\deg(U_B\xi_0) = \deg(U_B)$. Therefore

$$Q_{\mathrm{br}}[\psi_{U_B}] = \deg(U_B).$$

If U_B is topologically nontrivial, then its degree is nonzero, so the induced bridge charge is nonzero as well. □

Remark 7.8. *The induced charge is independent of the choice of reference doublet up to the natural $SU(2)$ redundancy, since different unit choices are related by the same diffeomorphic identification used above.*

Finite-energy transfer

The bridge matter functional is

$$L_m = -\frac{1}{2}K(\Phi)\, g^{\mu\nu}\nabla_\mu\bar{\psi}\,\nabla_\nu\psi - U(\psi,\Phi), \qquad U(\psi,\Phi) = \lambda_\psi\big(|\psi|^2 - v(\Phi)^2\big)^2. \tag{38}$$

For the canonical induced field,

$$|\psi_{U_B}|^2 = |v(\Phi)|^2\,|U_B\xi_0|^2 = v(\Phi)^2, \tag{39}$$

so the potential term vanishes identically:

$$U(\psi_{U_B},\Phi) = 0. \tag{40}$$

Differentiating,

$$\nabla_\mu\psi_{U_B} = (\partial_\mu v(\Phi))\,U_B\xi_0 + v(\Phi)\,(\nabla_\mu U_B)\xi_0. \tag{41}$$

Using unitarity and $\|\xi_0\| = 1$,

$$|\nabla\psi_{U_B}|^2 \le 2|\nabla v(\Phi)|^2 + 2|v(\Phi)|^2\,\|\nabla U_B\|^2. \tag{42}$$

Theorem 7.9 (Finite-energy transfer theorem)**.** *Assume, in addition to the hypotheses above, that on the post-threshold region:*

1. *$K(\Phi)$ is bounded above and below by positive constants;*
2. *$v(\Phi)$ is bounded;*
3. *$\nabla v(\Phi) \in L^2$ on the post-threshold spatial slice;*
4. *the frozen DTFE branch has finite gradient content on that slice, equivalently*

$$\int_\Sigma \|\nabla U_B\|^2\, d\mu_g < \infty. \tag{43}$$

Then the induced bridge field has finite bridge energy:

$$E[\psi_{U_B}] < \infty. \tag{44}$$

Proof. The potential term vanishes identically. For the kinetic term, the estimate above gives

$$K(\Phi)|\nabla\psi_{U_B}|^2 \le 2K_+|\nabla v(\Phi)|^2 + 2K_+v_+^2\|\nabla U_B\|^2, \tag{45}$$

where K_+ and v_+ are upper bounds for $K(\Phi)$ and $|v(\Phi)|$. Both terms on the right are integrable by assumptions (3) and (4). Hence the bridge energy is finite. □

Remark 7.10. *The finite-gradient assumption on U_B is the key regularity input in the present result. Deriving it directly from the full parent-action energy structure is part of the next strengthening step.*

Derived curvature sourcing

The imported bridge theorem already proves that finite-energy nontrivial topological sectors of ψ act as matter-energy concentrations and source curvature through

$$F(\Phi)G_{\mu\nu} = \kappa T^{\text{eff}}_{\mu\nu}. \tag{46}$$

We may therefore identify the induced DTFE contribution directly from the bridge matter Lagrangian.

Proposition 7.11 (Induced bridge stress tensor)**.** *For the canonical induced field ψ_{U_B}, the bridge matter stress tensor is*

$$T^{(\psi_{U_B})}_{\mu\nu} = K(\Phi)\Big(\nabla_{(\mu}\bar{\psi}_{U_B}\nabla_{\nu)}\psi_{U_B} - \frac{1}{2}g_{\mu\nu}\, g^{\alpha\beta}\nabla_\alpha\bar{\psi}_{U_B}\nabla_\beta\psi_{U_B}\Big), \tag{47}$$

since the potential term vanishes identically on ψ_{U_B}.

Proof. This is the standard stress tensor obtained by variation of the bridge matter Lagrangian with respect to the metric. Because

$$U(\psi_{U_B}, \Phi) = 0,$$

no potential contribution remains. □

Theorem 7.12 (Derived curvature-sourcing theorem)**.** *Under the hypotheses of the finite-energy transfer theorem and the topological transfer theorem, the canonical induced field ψ_{U_B} defines a finite-energy nontrivial topological bridge matter sector. Consequently it contributes a finite persistent source term to the bridge equation, and the post-threshold field equations take the form*

$$F(\Phi)G_{\mu\nu} = \kappa\Big(T^{(\Phi)}_{\mu\nu} + T^{(\psi_{U_B})}_{\mu\nu} + T^{(\text{adm})}_{\mu\nu}\Big). \tag{48}$$

Proof. By the finite-energy transfer theorem,

$$E[\psi_{U_B}] < \infty.$$

By the topological transfer theorem,

$$Q_{\text{br}}[\psi_{U_B}] \neq 0$$

whenever the DTFE branch is topologically nontrivial. The bridge matter theorem then applies directly: finite-energy nontrivial topological sectors of ψ define persistent matter-energy concentrations and therefore contribute to the effective curvature source. Substituting the explicit induced matter stress tensor gives the stated sourced bridge equation. □

Corollary 7.13 (Closed matter–geometry seam under post-threshold regularity)**.** *Let U_B be a finite-energy nontrivial DTFE branch with vacuum limit at infinity, satisfying the admissible topological setting above, and let*

$$\psi_{U_B} = v(\Phi)\, U_B\xi_0 \tag{49}$$

be the canonical induced bridge field. Under the post-threshold regularity hypotheses used above, the DTFE branch determines:

1. *a canonically induced bridge matter field;*
2. *a nontrivial bridge topological charge equal to the DTFE degree;*

3. *finite bridge matter energy;*

4. *a finite persistent curvature source in the post-threshold bridge equations.*

Therefore the DTFE branch closes the matter–geometry seam in derived form within the post-threshold bridge regime.

8 Strengthening programme

At this stage the unification problem is sharply defined.

Proposition 8.1 (Current unification seam)**.** *The remaining central task in Darke Theory quantum gravity–general relativity unification is to extend, relax, or remove the explicit post-threshold regularity assumptions under which the DTFE branch now enters the post-threshold relativistic sector in derived form.*

The immediate strengthening programme is therefore clear.

8.1 Strengthening of the finite-energy transfer

One may seek to derive the finite gradient content of the induced DTFE sector directly from the full parent-action energy structure rather than assuming it in bridge form.

8.2 Extension beyond the present post-threshold regularity class

One may seek to weaken the boundedness and integrability hypotheses on $K(\Phi)$, $v(\Phi)$, and $\nabla v(\Phi)$ while preserving the same transfer result.

8.3 Expanded weak-field and phenomenological extraction

Since the induced matter sector now enters the post-threshold curvature source in derived form, the next natural task is to extract explicit weak-field corrections and first phenomenological consequences from the sourced bridge equations.

8.4 Beyond the present matter candidate

The present paper identifies DTFE as the first concrete executable matter-sector route into the bridge. Future work may enlarge the matter catalogue without weakening the derived result established here.

9 Derived bridge theorem

We may now state the central theorem of the paper.

Theorem 9.1 (Derived DTFE bridge theorem)**.** *Assume:*

(i) the Darke Theory pre-geometric threshold structure, with geometry admissible only for $\Phi > \Phi_{\mathrm{crit}}$*;*

(ii) the relativistic closure–curvature bridge with post-threshold fields $(g_{\mu\nu}, \Phi, \psi)$, positive activation, hyperbolicity, finite admissibility, and the bridge matter theorem;

(iii) a finite-energy nontrivial DTFE branch U_B with vacuum limit at spatial infinity, satisfying the admissible topological setting above;

(iv) the post-threshold regularity hypotheses used in the finite-energy transfer theorem.

Then:

(a) below threshold, the theory admits a non-geometric coherence regime ordinarily described quantum mechanically;

(b) above threshold, the same theory admits an effective Lorentzian curvature regime sourced by persistent topological matter;

(c) the DTFE branch determines a canonical induced bridge matter field

$$\psi_{U_B} = v(\Phi)\, U_B \xi_0;$$

(d) this induced field has nontrivial bridge topological charge equal to the DTFE degree;

(e) this induced field has finite bridge energy;

(f) this induced field contributes a finite persistent source term to the post-threshold bridge equation.

Consequently, Darke Theory contains a derived concrete matter-sector bridge linking the DTFE topological branch to the post-threshold relativistic sector.

Proof. Item (a) is the threshold-separated consequence of the Darke Theory foundation below geometric admissibility. Item (b) follows from the imported relativistic closure–curvature bridge in the post-threshold regime. Item (c) is the canonical induced-field theorem. Item (d) is the topological transfer theorem. Item (e) is the finite-energy transfer theorem. Item (f) is the derived curvature-sourcing theorem. Therefore the same pre-geometric theory supports a non-geometric near-threshold regime and an effective Lorentzian post-threshold regime linked by a derived finite-energy nontrivial topological matter sector. □

Remark 9.2. *The decisive step is the closure of the matter–geometry seam in derived form for the first concrete matter candidate. The bridge is no longer merely structural. It now contains an explicit matter-sector realisation showing what passes from DTFE into the bridge and under which post-threshold regularity conditions that passage succeeds.*

10 Conclusion

Darke Theory places both the regime ordinarily described quantum mechanically and the regime ordinarily described relativistically within one pre-geometric threshold architecture. Below closure threshold, the theory places physics in a non-geometric coherence regime in which spacetime relations are not yet admissible. Above closure threshold, it admits an effective Lorentzian curvature regime governed by the relativistic closure–curvature bridge. The DTFE sector supplies the first concrete executable matter candidate for that bridge by providing a finite-energy nontrivial topological branch from which a canonical post-threshold matter field may be constructed.

The present paper derives the first concrete matter-sector seam in that architecture. It states the canonical induced bridge field, proves transfer of topological nontriviality, proves finite bridge matter energy under explicit post-threshold regularity assumptions, derives the corresponding induced stress tensor, and shows that the induced DTFE sector enters the post-threshold curvature equation as a finite persistent source.

The remaining task is now clear: strengthen the regularity side of the transfer, extend the result beyond the present hypothesis class, and extract weak-field and phenomenological consequences from the sourced bridge equations.

Darke Theory therefore contains a pre-geometric architecture underlying both quantum and relativistic regimes, and the DTFE matter sector furnishes a derived concrete bridge by which a nontrivial topological closure branch enters the post-threshold relativistic sector as finite matter and curvature source.

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